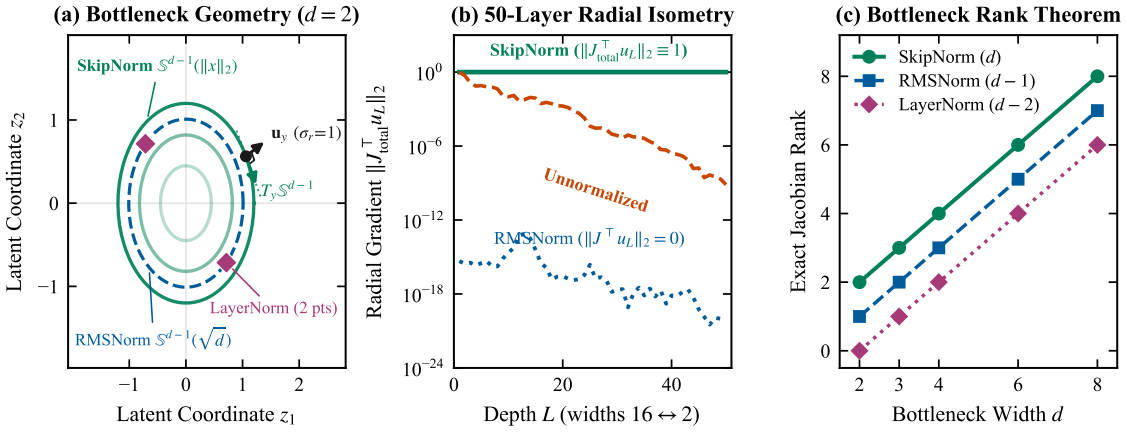

(a) Bottleneck Geometry ($d = 2$)
SkipNorm $\mathbb{S}^{d-1}(\|x\|_2)$
$\mathbf{u}_y$ (σ_r=1)
$T_y\mathbb{S}^{d-1}$
LayerNorm (2 pts)
RMSNorm $\mathbb{S}^{d-1}(\sqrt{d})$
Latent Coordinate z_1
Latent Coordinate z_2
(b) 50-Layer Radial Isometry
SkipNorm ($\|J_{\text{total}}^{\top} u_L\|_2 \equiv 1$)
Unnormalized
RMSNorm ($\|J^{\top} u_L\|_2 = 0$)
Radial Gradient $\|J_{\text{total}}^{\top} u_L\|_2$
Depth L (widths $16 \leftrightarrow 2$)
(c) Bottleneck Rank Theorem
SkipNorm (d)
RMSNorm ($d-1$)
LayerNorm ($d-2$)
Exact Jacobian Rank
Bottleneck Width d